6. An input of $25t^3u(t)$ is applied to the input of a Type 3 unity feedback system, as shown in Figure P7.1,

where

$$G(s) = \frac{210(s+4)(s+6)(s+11)(s+13)}{s^3(s+7)(s+14)(s+19)}$$

Find the steady-state error in position. [Section: 7.3]

$t^n \leftrightarrow \frac{n!}{s^{n+1}}$ $\qquad E(s) = \frac{R(s)}{1+G(s)}$

$25 \cdot t^3 \leftrightarrow \frac{6}{s^4} \cdot 25 = \frac{150}{s^4}$

$$\lim_{s\to 0} sE(s) = \frac{\frac{150}{s^3}}{\frac{210(s+4)(s+6)(s+11)(s+13)}{s^3(s+7)(s+14)(s+19)}} = \frac{150}{\frac{(210)(4)(6)(11)(13)}{(7)(14)(19)}}$$

$= 0,38$

12. For the unity feedback system of Figure P7.1, where

$$G(s) = \frac{K(s+2)(s+4)(s+6)}{s^2(s+5)(s+7)}$$

find the value of K to yield a static error constant of 10,000. [Section: 7.4]

$E(s) = \frac{R(s)}{1+G(s)}$

$R(s) = 1/s$

$\Rightarrow e_{ss} = \lim_{s\to 0} \frac{1}{1+\underbrace{G(s)}_{k_p}}$

$k_p = \lim_{s\to 0} G(s)$

$k_a = \lim_{s\to 0} s^2 G(s) = \lim_{s\to 0} \frac{k(s+2)(s+4)(s+6)}{s^2(s+5)(s+7)}$

$10\,000 = \frac{k \cdot 2 \cdot 4 \cdot 6}{5 \cdot 7}$

$\Rightarrow 7291 = k$

13. For the system shown in Figure P7.4, [Section: 7.3]

a. Find K_p, K_v, and K_a.

b. Find the steady-state error for an input of $50u(t)$, $50tu(t)$, and $50t^2u(t)$.

c. State the system type.

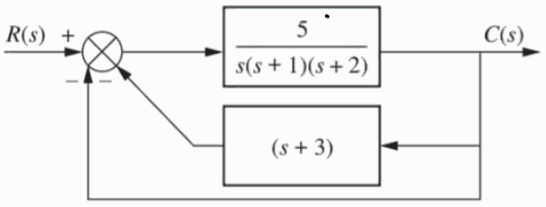


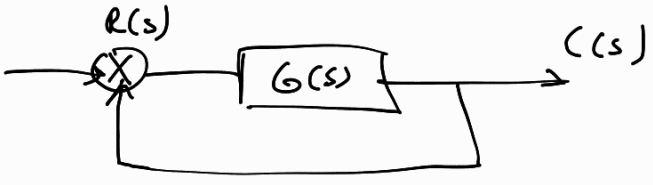
FIGURE P7.4

$$\frac{\frac{5}{s(s+1)(s+2)}}{1+\frac{5(s+3)}{s(s+1)(s+2)}} = \frac{5}{s(s+1)(s+2)+5(s+3)}$$

$$\frac{s(s+1)(s+2)+5(s+3)}{s(s+1)(s+2)}$$

$s(s+1)(s+2) = s^2+3s+2$

$$= \frac{5}{s^3+3s^2+2s+5s+15} = \frac{5}{s^3+3s^2+7s+15} = G(s)$$



$$K_p = \lim_{s\to 0} G(s) = \frac{1}{3}$$

$K_a = 0$

$K_a = 0$ for

b) $E_{ss} = \frac{50}{1+K_p} = \frac{50}{1+\frac{1}{3}} = 37.5$

for $50\,tu(t)$

$$E_{ss} = \lim_{s\to 0} \frac{50}{s} = \infty$$

for $50\,t^2u(t) \Rightarrow E_{ss} = \infty$

15. Find the system type for the system of Figure P7.5. [Section: 7.3]

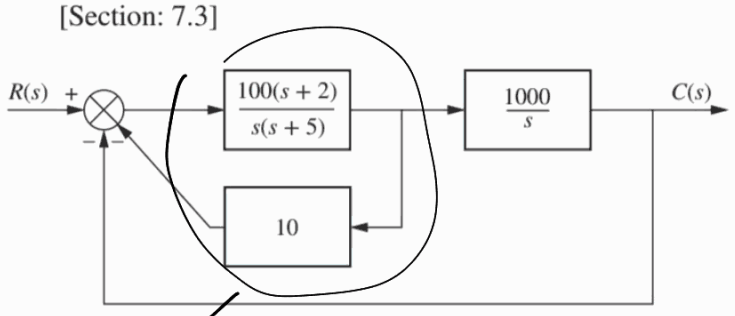


FIGURE P7.5

$$G(s) = \frac{\frac{100(s+2)}{s(s+5)}}{1 + \frac{1000(s+2)}{s(s+5)}} = \frac{100(s+2)}{s^2 + 5s + 1000s + 2000}$$

$$1005s$$

$$K_p = \lim_{s \to 0} \frac{100(s+2)}{s^2 + 1000s + 2000}\ (1000)$$

16. What are the restrictions on the feedforward transfer function $G_2(s)$ in the system of Figure P7.6 to obtain zero steady-state error for step inputs if: [Section: 7.3]

a. $G_1(s)$ is a Type 0 transfer function;

b. $G_1(s)$ is a Type 1 transfer function;

c. $G_1(s)$ is a Type 2 transfer function?

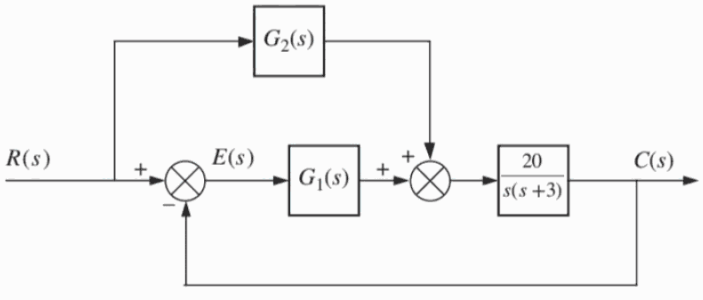
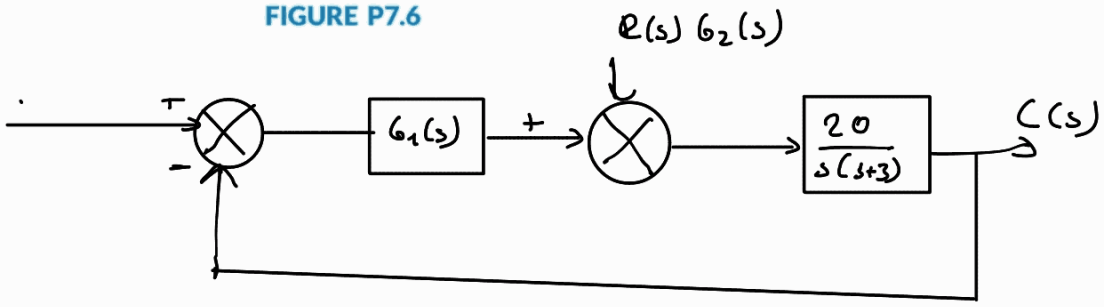


FIGURE P7.6



$$G(s) = (G_1 + R G_2)\left(\frac{20}{s(s+3)}\right)$$

$$\frac{1}{(s+3)} \qquad \frac{1}{s}$$

Type 1

22. The unity feedback system of Figure P7.1, where

$$G(s) = \frac{K(s^2 + 3s + 30)}{s^n(s + 5)}$$

is to have 1/6000 error between an input of $10tu(t)$ and the output in the steady state. [Section: 7.4]

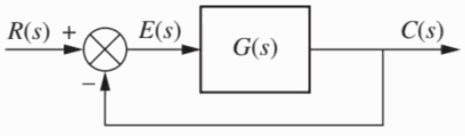


FIGURE P7.1

a. Find K and n to meet the specification.

b. What are K_p, K_v, and K_a?

$$\lim_{s \to 0} \frac{10}{s + sG(s)} = \frac{1}{6000}$$

$sG(s)$ → k_v

$n = 1$

$$\lim_{s \to 0} = \frac{s\,k(s^2 + 3s + 30)}{s(s+5)} = \frac{30k}{5}$$

$$\lim_{s \to 0} \frac{10}{s + \frac{30k}{6}} = \frac{1}{6000}$$

$$k_p = \infty \quad \Rightarrow \quad k = 10\,000$$

$k_v = 6k = 60\,000$

$k_a = 0$

27. The unity feedback system of Figure P7.1,where

$$G(s) = \frac{K(s+\alpha)}{(s+\beta)^2}$$

is to be designed to meet the following specifications: steady-state error for a unit step input = 0.1; damping ratio = 0.5; natural frequency = $\sqrt{10}$. Find K, α, and β. [Section: 7.4]

$\frac{\omega_n^2}{s^2+2\zeta\omega_n+\omega_n^2}$

$\frac{10}{s^2+\sqrt{10}+10}$

$e(\infty) = \frac{1}{1+k_p} = 0,1$

$k_p = \lim_{s\to 0} G(s) = \frac{k\alpha}{\beta^2}$

$\zeta = 0,5 \quad \omega_n = \sqrt{10}$

$10 = 1+k_p \quad k_p = 9 \quad \frac{k\alpha}{\beta^2} = 9$

32. Given the unity feedback control system of Figure P7.1, where

$$G(s) = \frac{K}{s(s+a)} =$$

$\frac{\frac{K}{s(s+a)}}{1+\frac{K}{s(s+a)}}$

find the following: [Section: 7.4]

a. K and a to yield $K_v = 1000$ and a 20% overshoot

b. K and a to yield a 1% error in the steady state and a 10% overshoot.

$= \frac{K}{s(s+a)+K} = \frac{K}{s^2+as+K}$ (ω_n^2 under K)

$M_p = e^{-t_p\sigma} \quad \sigma = \zeta\omega_n$

$= e^{-\frac{\pi}{\omega_d}\sigma} \quad t_p = \frac{\pi}{\omega_d}$

$\omega_d = \omega_n\sqrt{1-\zeta^2}$

$2\zeta\omega_n = a \quad \omega_n = \sqrt{K}$

$k_v \lim_{s\to 0} sG(s) = \frac{sK}{s(s+a)} = \frac{K}{a} = 1000$

$2\zeta\sqrt{K} = a$

$\zeta = \frac{2}{2\sqrt{K}}$

$M_p = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}} = 0,2$

$\frac{\omega_n^2}{2\zeta\omega_n} = 1000$

$\zeta = 0,45$

$\omega_n = 2000\zeta$

$\omega_n = 2000\zeta = 912$

$k = \omega_n^2 = 831\,744$

$\frac{K}{a} = 1000$

$\Rightarrow a = 831,744$

39. Design the values of K_1 and K_2 in the system of Figure P7.15 to meet the following specifications: Steady-state error component due to a unit step disturbance is −0.00001; steady-state error component due to a unit ramp input is 0.002. [Section: 7.5]

$e_D(\infty) = -0,000012$

$= -\frac{3}{2k_1}$

$\Rightarrow k_1 = 125000$

$\frac{1}{2k_1} = 0,00001$

$k_1 = \frac{1}{2.0,00001}$

$k_1 = 5000$

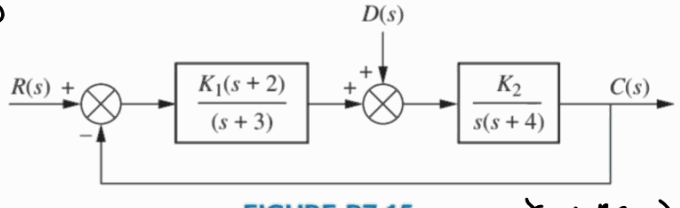
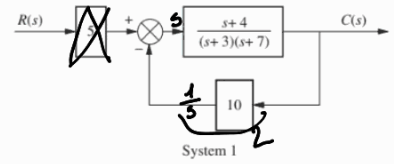


FIGURE P7.15

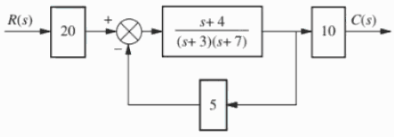
$e_D(\infty) = -\lim_{s\to 0} \frac{sG_2D}{1+G_1G_2} = -\lim_{s\to 0} \frac{s\left(\frac{k_2}{s(s+4)}\right)\frac{1}{s}}{1+\frac{k_1(s+2)}{s+3}\cdot\frac{k_2}{s(s+4)}} = \frac{\frac{k_2}{s(s+4)}}{\frac{s(s+3)(s+4)+k_1(s+2)k_2}{s(s+3)(s+4)}} = \frac{k_2(s+3)}{2k_1k_2} = -\frac{3}{2k_1}$

$e_R(\infty) = 0,002 = \frac{6}{k_1k_2} \Rightarrow k_2 = 9,6$

43. For each system shown in Figure P7.18, find the appropriate static error constant as well as the steady-state error, $r(\infty) - c(\infty)$, for unit step, ramp, and parabolic inputs. [Section: 7.6]



System 1



System 2

FIGURE P7.18

45. For the system shown in Figure P7.20, [Section: 7.6]

a. What is the system type?

b. What is the appropriate static error constant?

c. What is the value of the appropriate static error constant?

d. What is the steady-state error for a unit step input?

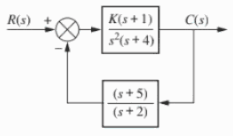


FIGURE P7.20

47. A dynamic voltage restorer (DVR) is a device that is connected in series to a power supply. It continuously monitors the voltage delivered to the load, and compensates voltage sags by applying the necessary extra voltage to maintain the load voltage constant.

In the model shown in Figure P7.22, u_r represents the desired reference voltage, u_o is the output voltage, and Z_L is the load impedance. All other parameters are internal to the DVR (*Lam, 2004*).

a. Assuming $Z_L = \dfrac{1}{sC_L}$, and $\beta \neq 1$, find the system's type.

b. Find the steady-state error to a unit step input as a function of β.

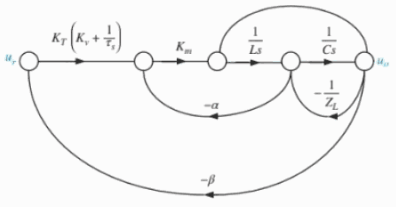


FIGURE P7.22 DVR Model

51. a. Show that the sensitivity to plant changes in the system of Figure P7.13 is

$$S_{T:P} = \frac{P}{T}\frac{\delta T}{\delta P} = \frac{1}{1 + L(s)}$$

where $L(s) = G(s)P(s)H(s)$ and

$$T(s) = \frac{C(s)}{R(s)} = \frac{F(s)}{H(s)} \cdot \frac{L(s)}{1 + L(s)}.$$

b. Show that $S_{T:P}(s) + \dfrac{T(s)H(s)}{F(s)} = 1$ for all values of s.

b. Show that $S_{T:P}(s) + \frac{F(s)}{} = 1$ for all values of s.

52. In Figure P7.13, $P(s) = \dfrac{2}{s}$,

$$T(s) = \frac{C(s)}{R(s)} = \frac{14K}{(s+1)(s+2)(s^2+5s+14)},$$

and

$$S_{T:P} = \frac{P}{T}\frac{\delta T}{\delta P} = \frac{s^2+5s}{s^2+5s+14}.$$

a. Find $F(s)$ and $G(s)$.

b. Find the value of K that will result in zero steady-state error for a unit step input.

58. As part of the development of a textile cross-lapper machine (*Kuo, 2010*), a torque input, $u(t) = \begin{cases} 1 & 0 \le t < 50 \\ -1 & 50 \le t < 100 \end{cases}$, is applied to the motor of one of the movable racks embedded in a feedback loop. The corresponding velocity output response is shown in Figure P7.27.

a. What is the open-loop system's type?

b. What is the steady-state error?

c. What would be the steady-state error for a ramp input?

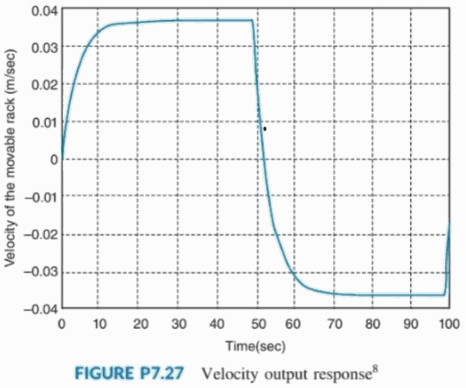


FIGURE P7.27 Velocity output response[8]

59. The block diagram in Figure P7.28 represents a motor driven by an amplifier with double-nested tachometer feedback loops (*Mitchell, 2010*).

a. Find the steady-state error of this system to a step input.

b. What is the system type?

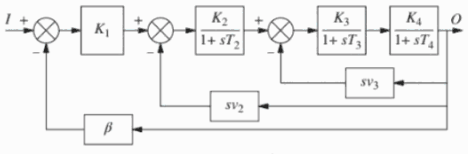


FIGURE P7.28[9]